

IPsec VPN Technical & Forensic Audit Report

Target File: ipsec_ikev2_web_browsing.pcap | Session: sess_20260901_53fdbb | Timestamp: 2026-09-01T12:09:32

1. Capture & Tunnel Forensics

- Packets Analyzed: 292 (IKE: 292, ESP: 0, Other: 0)
- Captured Duration: 58.53s | Size: 286.9 KB
- Protocol: IPsec (IKEv2) | Mode: Tunnel Mode

2. Cryptographic Security Assessment

- Encryption Cipher: 3DES-CBC
- Integrity / Authentication: HMAC-MD5
- Diffie-Hellman Group: MODP-1024 (Group 2)
- Perfect Forward Secrecy: Enabled | Replay Protection: Enabled

3. AI Traffic Classification

- Predicted Application: File Transfer (84% Confidence)
- Overall Security Score: 59/100 (CRITICAL) | Risk: 61/100 (HIGH)

4. Detected Threat Findings

- [HIGH] Weak Diffie-Hellman Group: Upgrade DH Group immediately to MODP-2048 or ECP-384
- [LOW] CBC Cipher Mode Utilization: Transition to AEAD mode such as AES-128-GCM or AES-256-GCM
- [INFO] Unencrypted Endpoint IP Metadata: Consider routing traffic through an anonymizing gateway or overlay if required